

PROFILE

Postdoctoral researcher and senior lecturer working at the intersection of *immersive technology, software product engineering, and pedagogy*. I lead research-informed teaching across postgraduate programmes and run mixed-methods user studies on multiplayer location-based AR, spatial interaction, and player experience — in partnership with Sony Interactive Entertainment, Niantic, and the University of Canterbury. My practice is built around experiential learning, industry collaboration, and values-driven, inclusive learning environments.

01

Work

Jun 2025 — Present

HIT LAB NZ · CHRISTCHURCH

Postdoctoral Researcher

University of Canterbury — Human Interface Technology Lab NZ

An interdisciplinary research centre for human-computer interaction, immersive technologies, and user-experience innovation.

- Lead applied XR research in partnership with **Sony Interactive Entertainment (SIE), Japan**, spanning design, implementation, and evaluation of VR applications.
- Run mixed-methods studies on user experience, comfort, and usability in immersive environments.
- Design multi-modal experimental platforms integrating XR Interaction Toolkit and OpenXR.
- Mentor postgraduate students in experimental design, Unity development, and publication writing.

Feb 2023 — Present

YOOBEE · CHRISTCHURCH

Senior Lecturer

Yoobee College of Creative Innovation

A leading NZ tertiary institute delivering creative technology and business programmes.

- Independently designed and built an **AI-first, learner-friendly interactive LMS** for blended postgraduate delivery — video tutorials, gamified lessons and challenges, AI-driven assessments, and a business-intelligence dashboard giving the lecturer real-time cohort analytics.
- Teach masters-level courses: Strategic Information Systems Planning, Database Management, and IT Project Management.
- Supervise capstone projects; coordinate teaching assistants; lead moderation and quality assurance aligned with NZQA standards.
- Recognised for innovative, high-engagement, research-informed teaching methods.

Jan 2020 — Sep 2022

OPEN.UOM.LK · NATIONAL MOOC

Platform Lead — Software Engineering Strand

Open Learning Platform, University of Moratuwa · Sri Lanka

Sri Lanka's first large-scale MOOC platform — free, self-paced courses for IT and engineering learners.

- Co-designed, configured, and deployed the platform end-to-end; scaled to **150,000+ registered learners**.
- Coordinated cross-faculty collaboration to ship a unified Software Engineering curriculum.
- Built fully automated registration, grading, and certification — enabling scalable delivery with minimal manual ops.
- Recognised by the university as a flagship open-education project.

Lecturer

University of Moratuwa — Faculty of Information Technology

- Taught Database Systems, HCI, Software Engineering, and Enterprise Application Development at UG and PG level.
- Designed an AR/VR learning framework subsequently adopted for postgraduate teaching.
- Contributed to curriculum development for HCI and enterprise development modules.

Technical Lead & Senior Software Engineer

99X Technology — Sri Lanka

Award-winning product engineering company delivering software to global clients.

- Led engineering teams building large-scale enterprise applications; promoted to Technical Lead.
- Introduced AR-based features and embedded Agile development practices across teams.
- Ran technical training and mentored junior engineers; multiple Best-Employee awards.

02

Education & training

Doctor of Philosophy – Human Interface Technology

University of Canterbury, Aotearoa New Zealand

Thesis: Designing multiplayer location-based augmented reality games that connect remote players and places. ir.canterbury.ac.nz ↗

- Three mixed-methods user studies combining AR, eye-tracking, and player-experience metrics.
- Produced multiple peer-reviewed publications and design guidelines for immersive game interaction.
- Fully funded under the Applied Immersive Gaming Initiative (AIGI).

Specialisation in Virtual Reality

Goldsmiths, University of London

BSc (Hons) in Information Technology

University of Moratuwa, Sri Lanka — Faculty of Information Technology

03

Selected publications

2025 JOURNAL

Wickramasinghe, Y. S., Lukosch, H. K., Everett, J., & Lukosch, S. (2025). *Evaluating spatial decision-making and player experience in a remote multiplayer augmented reality hide-and-seek game*. Multimodal Technologies and Interaction, 9(8), 79. doi.org/10.3390/mti9080079

2025 OZCHI

Wickramasinghe, Y. S., Lukosch, H., Everett, J., & Lukosch, S. (2025). *On the impact of augmented reality game mechanics on the player experience in remote multiplayer gameplay*. In Proceedings of the 37th Australian Conference on Human-Computer Interaction (OzCHI '25) (pp. 426–439). Association for Computing Machinery. doi.org/10.1145/3764687.3764699

2025 JOURNAL

Wickramasinghe, Y. S., Lukosch, H. K., Everett, J., & Lukosch, S. (2025). *Representing remote locations with location-based augmented reality game design*. Entertainment Computing, 53, 100932. doi.org/10.1016/j.entcom.2025.100932

2025 IEEE VR

Wickramasinghe, Y. S., Lukosch, H., Everett, J., & Lukosch, S. (2025). *Augmented hide-and-seek: evaluating spatial decision-making and player experience in a multiplayer location-based game*. IEEE VR & 3D User Interfaces Abstracts & Workshops (VRW 2025), Saint-Malo, France, pp. 1228–1229. doi.org/10.1109/VRW66409.2025.00260

2024	ACM SUI	Ritter, M., Liew, K., Wickramasinghe, Y. S., & Lukosch, S. (2024). <i>Trustful trading in shared augmented reality</i> . Proceedings of the 2024 ACM Symposium on Spatial User Interaction (SUI '24), Article 4, 1–12. doi.org/10.1145/3677386.3682086
2023	ISAGA	Wickramasinghe, Y. S., Lukosch, S., & Lukosch, H. (2023). <i>Designing immersive multiplayer location-based augmented reality games with remotely shared spaces</i> . In Becu, N. (Ed.), Simulation and Gaming for Social and Environmental Transitions — Proceedings of the 54th ISAGA Conference (pp. 204–214). halshs-04209935
2022	KDU	Inparajah, J., & Wickramasinghe, Y. S. (2022). <i>A review on reimagining medical education with virtual reality in emerging medical disciplines</i> . Proceedings of the 15th International Research Conference, KDU, Sri Lanka. ir.kdu.ac.lk/handle/345/6503
2017	IEEE	Wickramasinghe, Y. S., De Saram, P. S. R. S., Liyanage, C. P., Rangika, L. N. R., & Ranathunga, L. (2017). <i>Virtual reality markup framework for generating interactive indoor environment</i> . IEEE 3rd Int. Conf. on Engineering Technologies and Social Sciences (ICETSS 2017), Bangkok, pp. 1–6. doi.org/10.1109/ICETSS.2017.8324175
2015	THESIS	Wickramasinghe, Y. S., & Gunasekara, E. H. (2015). <i>A mobile application with augmented reality to enhance Sinhala learning experience for children</i> . University of Moratuwa Digital Library. dl.lib.mrt.ac.lk

Teaching & supervision

POSTGRADUATE SUPERVISION	2023 – Now	Kotuwegedara, H. — Master's Thesis, Yoobee Colleges · <i>Primary supervisor</i> . Prompting for Security: LLM Code Generation in Angular.
	2022 – 2024	Ritter, M. — Master's Thesis, HIT Lab NZ, University of Canterbury · <i>Co-supervisor</i> . Trust and trustworthiness while exchanging virtual items in shared augmented reality. doi.org/10.26021/15296
	2021 – 2022	Inparajah, J. — Final-Year Thesis, University of Moratuwa · <i>Primary supervisor</i> . Reimagining Medical Education with VR in Emerging Medical Disciplines.
COURSES DELIVERED	2025	MBI804 — IT Project Management (Master's) · Senior Lecturer, Yoobee Colleges
	2025	MBI802 — Database Management (Master's) · Senior Lecturer, Yoobee Colleges
	2025	MBI800 — Strategic Information Systems (Master's) · Senior Lecturer, Yoobee Colleges
	2023	HITD603 — Human Interface Technology (Master's) · Workshops on AR application development, HIT Lab NZ
	2019–22	IN1600 — Multimedia & Web Technologies (BSc) · Lecturer, University of Moratuwa
	2020–22	IN2100 — Object Oriented Programming (BSc) · Lecturer, University of Moratuwa

Invited talks

SELECTED TALKS	2024	Breaking the Wall of Loneliness through Play · Falling Walls Lab
	2023	The First AR Game Dev Rant — Panel with Evie, Melanie, Devi & Yasas · NZ Game Developers Conference
	2023	Designing Location-based AR Games Enhancing Spatial Presence · Invited talk, NZ Game Developers Conference

Grants & recognition

GRANTS & SCHOLARSHIPS	2024	Niantic Studio Developer Accelerator Grant · Niantic, Inc.
	2022–25	Applied Immersive Gaming Initiative (AIGI) Full Scholarship · University of Canterbury

2024	Winner — Technology Track, Napkin Challenge, University of Canterbury
2022	People's Choice Award, Visualize Your Thesis, University of Canterbury
2018–20	Excellence Awards — Best Initiative Team Lead, 99X (3 consecutive years)
2020	Emerging Xian Award — Top 3 Nominee, 99X
2015	1st Runner-up — Microsoft Imagine Cup National Finals, Sri Lanka
2015	Champion — Hackanix Inter-University Hackathon, Microsoft Sri Lanka & AIESEC
2015	1st Runner-up — Fuegoistica V2.0 Code Ultimate Hackathon, Sri Lanka
2016	2nd Runner-up — Dialog Ideamart "Mora Hack" · Finalist, SSAsia12 Code-A-Thon (IEEE) · Semi-finalist, DisruptX (SLASSCOM & IEEE)
2016	Google Translation Marathon, Google Sri Lanka

What I bring

PRACTICE

■ EdTech & Platform Building

Independently designed and built an **AI-first, learner-friendly LMS** at Yoobee — video tutorials, gamified lessons, AI-driven assessments, and a BI dashboard for live cohort insight. Earlier, co-built Sri Lanka's first national MOOC platform reaching **150k+ learners** with automated grading and certification.

■ Teaching & Curriculum Design

Research-led postgraduate and undergraduate courses across Information Systems, HCI, and Project Management. Built gamified, industry-connected learning tools that lifted engagement and outcomes.

■ Technical & Digital Expertise

Unity 3D, Niantic Lightship ARDK, ARFoundation, OpenXR, XR Interaction Toolkit. Data analytics in Python, R, and SQL. Prototype-to-production across research and teaching contexts.

■ Collaboration & Leadership

Multi-institutional team leadership across research and curriculum initiatives. Curriculum developer, conference speaker, and outreach mentor.

■ Research & Innovation

Peer-reviewed publications in HCI and AR; empirical user studies using mixed methods, eye-tracking, and player-experience metrics. Collaborations with Sony Interactive Entertainment and Niantic.

■ Communication & Mentorship

Supervise postgraduate research, mentor teaching assistants, and present at academic and industry conferences. Known for making complex technical ideas accessible.

■ Critical Thinking & Problem-Solving

Design-led reasoning applied to pedagogical and technological challenges, with a commitment to evidence-based, reflective practice.

Service & certifications

ACADEMIC SERVICE

2025	Paper Reviewer — ACM Symposium on Virtual Reality Software and Technology (VRST)
2024	Student Volunteer — IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)
2024	Student Volunteer — International Simulation and Gaming Association (ISAGA) Conference
2024–25	Postgraduate Student Mentor — University of Canterbury
2023 – Now	Blogger, <i>Emerging Technologies</i> — readclub.me & Medium. AR/VR & HCI for an international research and design audience.

CERTIFICATIONS & TRAINING

2025	AI Fluency for Educators · Anthropic — Certificate of Completion
–	Certificate in Teaching Higher Education · Best Portfolio Submission & Best Research Presentation Honours
–	Scrum Foundation Professional Certificate (SFPC)
–	Certificate in Using Eye Trackers in Research · Tobii Academy
–	GitLab Certified Associate
–	Online Presence and Digital Footprint · University of Edinburgh
–	Product Development: Customer Journey Mapping · Coursera